2.2 Well-Definedness, and the Arithmetic of the Integers

Section 2.1 built $\mathbb{Z}$ as the set of classes of $\sim$ on $\mathbb{N} \times \mathbb{N}$, and gave each integer a normal form (Proposition 2.1.2). The integers cannot yet be added. Every operation on classes will be defined by a formula on representatives, and *Sets and Functions*, Section 5.2 showed what can go wrong: a formula that gives two answers for one class (Sets and Functions, Example (A rule on classes that is not a function)). This section proves, once and for all, the lemma that says when such a formula defines a function. It then uses the lemma for the sum, product and negative of integers, and finds a copy of $\mathbb{N}$ inside $\mathbb{Z}$.

2.2.1 Definitions through representatives

In *Sets and Functions*, Section 5.2 the addition of $\mathbb{Z}/n$ was defined by $[a] + [b] \coloneqq [a + b]$, and Sets and Functions, Proposition 5.2.5 checked that the right side depends only on the classes, not on the integers a and b used to name them. In Sets and Functions, Example (A rule on classes that is not a function) the same kind of formula, $[a]_6 \mapsto [a]_4$, gave one class two values, and so defined no function.

The same situation will recur throughout this volume: for $\mathbb{Z}$ in this section, for $\mathbb{Q}$ in Section 3.1, and for $\mathbb{R}$ later in this volume. Each time, a set is a quotient $X/\!\sim$, and something is defined on it by a formula on elements of X. Rather than repeat the argument of Sets and Functions, Proposition 5.2.5 for every formula, we separate it into two parts: a general lemma, proved here once, which turns a **check on representatives** into a function on classes; and the check itself, which is different for each formula and is all that remains to do each time.

A rule "$[x] \mapsto$ something computed from x" is called **well defined** when representatives of the same class always give the same result. The lemma below says that being well defined is enough, in three settings: a map from one quotient to another, a binary operation on a quotient, and a property of classes.

2.2.2 The lemma

A rule on classes is a function as soon as equivalent inputs give equivalent outputs.

Theorem 2.2.1. Functions on Classes

Let $\sim$ be an equivalence relation on a set X, and $\approx$ an equivalence relation on a set Y. Classes are taken for $\sim$ in X and for $\approx$ in Y.

(a) Let $f\colon X \to Y$ be a function such that, **for all** $x, x' \in X$, $x \sim x'$ implies $f(x) \approx f(x')$. Then there is **exactly one** function $F\colon X/\!\sim \to Y/\!\approx$ with $F([x]) = [f(x)]$ for every $x \in X$.

(b) Let $g\colon X \times X \to X$ be a function such that, **for all** $x, x', y, y' \in X$,

$$\begin{aligned} x \sim x' &\implies g(x,y) \sim g(x',y), \\ y \sim y' &\implies g(x,y) \sim g(x,y'). \end{aligned}$$

Then there is **exactly one** function $G\colon (X/\!\sim) \times (X/\!\sim) \to X/\!\sim$ with $G([x],[y]) = [g(x,y)]$ for all $x, y \in X$.

(c) Let $S \subseteq X$ be such that, for all $x, x' \in X$, $x \sim x'$ and $x \in S$ imply $x' \in S$. Let $\bar{S} \coloneqq \{[x] : x \in S\} \subseteq X/\sim$. Then, **for every** $x \in X$, $[x] \in \bar{S}$ if and only if $x \in S$.

In words: (a) says that a map on elements that respects the two relations passes to a map on classes; (b) says that a binary operation that respects $\sim$ **in each argument separately** passes to an operation on classes; (c) says that a property that respects $\sim$ can be asked of a class, and the answer is the answer for any of its representatives. Clause (a) will give the negative of an integer, (b) its sum and product, and (c) the positive integers in Section 2.4.

Idea. Clause (a) is Sets and Functions, Example (Method: checking that a definition on classes is well defined) turned into a theorem: steps (1)–(3) of that method are its hypothesis, and step (4) is the graph built below. A function is a graph with **exactly one** output for each input (Sets and Functions, Definition 2.1.1). The obvious graph is the set of all pairs $([x], [f(x)])$. It has at least one output for each class, because every class is some $[x]$. The only danger is two outputs for one class, coming from two representatives x and x' of it; Sets and Functions, Theorem 5.2.1 (b) turns "same class" into $x \sim x'$, the hypothesis turns that into $f(x) \approx f(x')$, and Sets and Functions, Theorem 5.2.1 (b) turns it back into "same output". For (b) the same plan works, once the two one-argument hypotheses are chained into one statement about both arguments by transitivity.

Proof. (a) Let

$$G_F \coloneqq \{(C, D) \in (X/\sim) \times (Y/\approx) : C = [x] \text{ and } D = [f(x)] \text{ for some } x \in X\},$$

a set by specification (Sets and Functions, Definition 1.2.3). We check the two clauses of Sets and Functions, Definition 2.1.1. (Fn1): every $C \in X/\sim$ is $[x]$ for some $x \in X$ (Sets and Functions, Definition 5.2.4), and then $(C, [f(x)]) \in G_F$. (Fn2): suppose $(C, D) \in G_F$ and $(C, D') \in G_F$, say $C = [x]$, $D = [f(x)]$, and $C = [x']$, $D' = [f(x')]$. Then $[x] = [x']$, so $x \sim x'$ by Sets and Functions, Theorem 5.2.1 (b). By the hypothesis, $f(x) \approx f(x')$, so $[f(x)] = [f(x')]$ by Sets and Functions, Theorem 5.2.1 (b), that is, $D = D'$. So F, the function with graph G_F and codomain $Y/\approx$, is a function $X/\sim \to Y/\approx$, and $F([x]) = [f(x)]$ because $([x], [f(x)]) \in G_F$.

For uniqueness, let $F' \colon X/\sim \to Y/\approx$ also satisfy $F'([x]) = [f(x)]$ for every x. Every $C \in X/\sim$ is some $[x]$, and then $F'(C) = [f(x)] = F(C)$. The two functions have the same domain and codomain, so $F' = F$ by Sets and Functions, Proposition 2.1.1.

(b) First, for all $x, x', y, y' \in X$ with $x \sim x'$ and $y \sim y'$,

$$g(x, y) \sim g(x', y) \quad \text{and} \quad g(x', y) \sim g(x', y'),$$

by the first hypothesis (with y fixed) and the second (with x' fixed). By transitivity, (Eq3) of Sets and Functions, Definition 5.2.1, $g(x, y) \sim g(x', y')$.

The rest is the proof of (a) with pairs of classes in place of classes: the graph of all $(([x], [y]), [g(x, y)])$ has an output for each pair of classes, since each pair of classes is $([x], [y])$ for some x, y (Sets and Functions, Definition 1.4.2, Sets and Functions, Definition 5.2.4), and only one, by the paragraph above, Sets and Functions, Definition 1.4.1 (equal pairs of classes have equal entries) and Sets and Functions, Theorem 5.2.1 (b); so it defines a function G, and G is unique as in (a) (Sets and Functions, Proposition 2.1.1).

Second reading · *The graph of G, checked as in (a)*

Now let G_G be the set of $(P, E) \in ((X/\sim) \times (X/\sim)) \times (X/\sim)$ such that $P = ([x], [y])$ and $E = [g(x, y)]$ for some $x, y \in X$, a set by specification. (Fn1): an element of $(X/\sim) \times (X/\sim)$ is a pair (C, D) of classes (Sets and Functions, Definition 1.4.2), and $C = [x]$, $D = [y]$ for some $x, y \in X$ (Sets and Functions, Definition 5.2.4); then $((C, D), [g(x, y)]) \in G_G$. (Fn2): suppose (P, E) and (P, E') lie in G_G, with $P = ([x], [y])$, $E = [g(x, y)]$, and $P = ([x'], [y'])$, $E' = [g(x', y')]$. From $([x], [y]) = ([x'], [y'])$ the characteristic property of pairs (Sets and Functions, Definition 1.4.1) gives $[x] = [x']$ and $[y] = [y']$, so $x \sim x'$ and $y \sim y'$ by Sets and Functions, Theorem 5.2.1 (b). By the first paragraph, $g(x, y) \sim g(x', y')$, so $E = E'$ by Sets and Functions, Theorem 5.2.1 (b). So G_G, with codomain $X/\sim$, defines a function G with $G([x], [y]) = [g(x, y)]$. Uniqueness is as in (a), by Sets and Functions, Proposition 2.1.1.

(c) ($\Leftarrow$) If $x \in S$, then $[x] \in \bar{S}$ by the definition of $\bar{S}$. ($\Rightarrow$) If $[x] \in \bar{S}$, then $[x] = [x']$ for some $x' \in S$. By Sets and Functions, Theorem 5.2.1 (b), $x \sim x'$, so $x' \sim x$ by (Eq2). The hypothesis, applied to $x' \sim x$ and $x' \in S$, gives $x \in S$.

This proves the theorem.

■

The functions F and G are said to be **induced** by f and g. In practice we write the conclusion of (a) as "the rule $[x] \mapsto [f(x)]$ is well defined", and of (b) as "the operation $[x] * [y] := [g(x, y)]$ is well defined", and the work is always the check of the hypotheses.

Remark. **Each hypothesis has a job.** Without the hypothesis of (c) the conclusion fails. On $\mathbb{N} \times \mathbb{N}$ with the relation $\sim$ of Definition 2.1.1, let $S := \{(a, b) : 0 < a\}$. Then $(1, 1) \in S$, so $[(1, 1)] \in \bar{S}$; and $[(0, 0)] = [(1, 1)]$, so $[(0, 0)] \in \bar{S}$, while $(0, 0) \notin S$. Membership in $\bar{S}$ does not tell us whether a particular representative lies in S. The hypotheses of (a) and (b) are needed in the same way, and the examples below show them failing.

2.2.3 Sum, product and negative of integers

To guess the formulas, borrow the school picture once more. If (a, b) records "$a - b$" and (c, d) records "$c - d$", then school algebra gives

$$\begin{aligned} (a - b) + (c - d) &= (a + c) - (b + d), \\ (a - b)(c - d) &= (ac + bd) - (ad + bc), \\ -(a - b) &= b - a. \end{aligned}$$

These identities are a **heuristic**: they come from the school integers, which this chapter does not assume, and they are used only to find the definition. Each right side is again a difference of two natural numbers, so it can be recorded as a pair.

Definition 2.2.1. Sum, Product and Negative of Integers

For all $(a,b),(c,d) \in \mathbb{N}\times\mathbb{N}$, let

$$\begin{aligned}[(a,b)]+[(c,d)] &\coloneqq [(a+c,\ b+d)],\\ [(a,b)]\cdot[(c,d)] &\coloneqq [(ac+bd,\ ad+bc)],\\ -[(a,b)] &\coloneqq [(b,a)].\end{aligned}$$

The **sum** $+\colon \mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$ and **product** $\cdot\colon \mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$ are the functions that Theorem 2.2.1 (b), with $X=\mathbb{N}\times\mathbb{N}$ and the equivalence relation $\sim$ of Proposition 2.1.1, induces from the maps $g_+,\ g_\cdot\colon (\mathbb{N}\times\mathbb{N})\times(\mathbb{N}\times\mathbb{N})\to\mathbb{N}\times\mathbb{N}$ given by the pairs on the right; the **negative** $-\colon\mathbb{Z}\to\mathbb{Z}$ is the function that Theorem 2.2.1 (a) induces from $(a,b)\mapsto(b,a)$. We also write xy for $x\cdot y$.

In words: to add integers, add representatives coordinatewise; to multiply, use the cross formula; to negate, swap the coordinates. All the arithmetic on the right happens in $\mathbb{N}$, where it is always possible. The definition is legitimate only if the hypotheses of Theorem 2.2.1 hold for the three maps on pairs; the next proposition checks them, and until then the three formulas are rules, not yet functions.

The proof below uses the move this chapter is built on, after Sets and Functions, Example (Method: checking that a definition on classes is well defined):

Example. Method: check on representatives, one argument at a time

Template. To show that $[x]*[y]\coloneqq[g(x,y)]$ is well defined: (1) fix y, take $x\sim x'$, and derive $g(x,y)\sim g(x',y)$ from the one equation that $x\sim x'$ unwinds to; (2) fix x', take $y\sim y'$, and derive $g(x',y)\sim g(x',y')$; if $g(x,y)=g(y,x)$ always, (2) is (1) with the roles exchanged; (3) conclude by Theorem 2.2.1 (b). A rule on single classes needs only (1), and then clause (a).

When to use it. Every operation on a quotient given by a formula on representatives: the sum and product of $\mathbb{Z}$, and every later quotient built the same way.

The usual failure. Checking one numerical instance; using a law of $\mathbb{Z}$ to check an operation of $\mathbb{Z}$ (the computation stays in $\mathbb{N}$); claiming symmetry for a formula that lacks it, such as $[(a+d,\ b+c)]$.

Quick check. Does $[(a,b)]\mapsto[(a+a,\ b)]$ define a function $\mathbb{Z}\to\mathbb{Z}$?

Solution. No. $(0,0)\sim(1,1)$, since $0+1=0+1$, but the outputs are $(0+0,0)=(0,0)$ and $(1+1,1)=(2,1)$, and $(0,0)\not\sim(2,1)$, since $0+1\neq 0+2$. In school terms the rule computes $2a-b$, which is not a function of $a-b$.

Proposition 2.2.2. The Operations on Integers Are Well Defined

The maps g_+ and $g_\cdot$ of Definition 2.2.1 satisfy the hypotheses of Theorem 2.2.1 (b), and the map $(a,b)\mapsto(b,a)$ satisfies those of Theorem 2.2.1 (a), with $\sim$ on both sides. Hence the sum, product and negative of Definition 2.2.1 are well-defined functions.

Idea (Scratch work). The sum is one line; the product is the real check. Take the first argument (step (1) of the box):

Givens	Goal
$(a,b) \sim (a',b')$; (c,d) fixed	$g.((a,b),(c,d)) \sim g.((a',b'),(c,d))$
$a+b' = b+a'$	$(ac+bd)+(a'd+b'c) = (ad+bc)+(a'c+b'd)$
every term of the goal contains c or d; the given contains neither	multiply the given by c, and, read from right to left, by d
$ac+b'c = bc+a'c$ and $bd+a'd = ad+b'd$	add the two and regroup: the goal

Both formulas are unchanged when the two input pairs are exchanged, so step (2) comes free.

Before reading on: which hypothesis of Theorem 2.2.1 (b) does this table check?

Proof. Let $(a,b),(a',b'),(c,d) \in \mathbb{N}\times\mathbb{N}$ with $(a,b)\sim(a',b')$, that is,

$$a+b' = b+a'. \tag{2.2.1}$$

All rearrangements below use Theorem 1.2.3: commutativity and associativity of $+$, commutativity of $\cdot$, and distributivity.

Sum, first argument. We need $(a+c,b+d)\sim(a'+c,b'+d)$, that is, $(a+c)+(b'+d) = (b+d)+(a'+c)$. The left side is $(a+b')+(c+d)$, which by Eq. 2.2.1 equals $(b+a')+(c+d)$, and that is the right side rearranged.

Product, first argument. We need

$$(ac+bd,\ ad+bc) \sim (a'c+b'd,\ a'd+b'c),$$

that is, the cross-sum equation

$$(ac+bd)+(a'd+b'c) = (ad+bc)+(a'c+b'd). \tag{2.2.2}$$

Multiply both sides of Eq. 2.2.1 by c and use distributivity: $ac+b'c = bc+a'c$. Multiply both sides of Eq. 2.2.1, read from right to left, by d: $bd+a'd = ad+b'd$. Adding these two equations,

$$(ac+b'c)+(bd+a'd) = (bc+a'c)+(ad+b'd).$$

The left side rearranges to the left side of Eq. 2.2.2, and the right side to its right side. This proves Eq. 2.2.2.

Second arguments. By commutativity in $\mathbb{N}$, $g_+((a,b),(c,d)) = (a+c,b+d) = (c+a,d+b) = g_+((c,d),(a,b))$, and likewise

$$g.((a,b),(c,d)) = (ac+bd,\ ad+bc) = (ca+db,\ cb+da) = g.((c,d),(a,b)).$$

So if $(c,d) \sim (c',d')$, then, by the first-argument case applied with the roles of the two pairs exchanged,

$$g((a,b),(c,d)) = g((c,d),(a,b)) \sim g((c',d'),(a,b)) = g((a,b),(c',d'))$$

for $g = g_+$ and $g = g_\cdot$. This is the second hypothesis of Theorem 2.2.1 (b).

Negative. The pair (b,a) is related to (b',a') when $b + a' = a + b'$, and that is Eq. 2.2.1 read from right to left. This is the hypothesis of Theorem 2.2.1 (a).

By Theorem 2.2.1 (b) and (a), the three rules define functions $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ and $\mathbb{Z} \to \mathbb{Z}$, as claimed.

■

The table checks the first hypothesis of Theorem 2.2.1 (b), $g(x,y) \sim g(x',y)$, for the product; symmetry gives the second, and the first paragraph of that proof joins the two by transitivity of $\sim$. Changing both at once is also possible, but the computation then mixes four equations in eight unknowns; one argument at a time keeps each computation to a single hypothesis.

2.2.4 Rules that fail

Every check happens in the hypotheses of the lemma. Here they fail, next to a rule where they hold.

Example. The first coordinate is not a function of the class

(a) Does the rule $[(a,b)] \mapsto a$ define a function $\mathbb{Z} \to \mathbb{N}$?

(b) Does the rule $[(a,b)] \mapsto [(a+1,b)]$ define a function $\mathbb{Z} \to \mathbb{Z}$?

(c) A student argues: "Each integer x has a pair with least first coordinate, by Theorem 1.3.3 applied to the non-empty set of first coordinates of the pairs in x; call that least first coordinate $h(x)$. Then h is a function $\mathbb{Z} \to \mathbb{N}$, since each x determines its own least first coordinate. So $[(a,b)] \mapsto a$ is a function, namely h, with $h([(a,b)]) = a$ for every (a,b)." Find the error.

Solution.

(a) No. $(0,2) \sim (1,3)$, since $0 + 3 = 2 + 1$, so $[(0,2)] = [(1,3)]$ by Sets and Functions, Theorem 5.2.1 (b). Read with the representative $(0,2)$, the rule gives 0; read with $(1,3)$, it gives 1. One input gets two values, and (Fn2) of Sets and Functions, Definition 2.1.1 fails. The hypothesis of Theorem 2.2.1 (a), with $f(a,b) := a$ and equality on $\mathbb{N}$ as $\approx$ (so that $f(x) \approx f(x')$ means $f(x) = f(x')$), fails at $x = (0,2)$, $x' = (1,3)$.

(b) Yes. Let $(a,b) \sim (a',b')$, so $a + b' = b + a'$. Then

$$(a+1) + b' = (a + b') + 1 = (b + a') + 1 = b + (a' + 1)$$

by Theorem 1.2.3 (b), (c), which is $(a+1,b) \sim (a'+1,b')$. By Theorem 2.2.1 (a), the rule is a function. In fact it is $x \mapsto x + [(1,0)]$, since $[(a,b)] + [(1,0)] = [(a+1,b+0)] = [(a+1,b)]$.

(c) The conclusion is false, by (a), so some step fails. The function h is fine: the least first coordinate is computed from the set x alone. The failing step is the last clause,

"$h([(a,b)]) = a$ for every (a,b)". It holds only when (a,b) happens to be the pair with least first coordinate. The argument chose a particular representative to define h, and then used an **arbitrary** one. Counterexample to the step: $x = [(1,3)] = [(0,2)]$ contains $(0,2)$, so $h(x) = 0$, while the representative $(1,3)$ has first coordinate 1.

Warning. **A formula that looks right need not be well defined.** "$[(a,b)] \cdot [(c,d)] \coloneqq [(ac,bd)]$, like the sum" is not a definition: $(1,0) \sim (2,1)$, since $1+1 = 0+2$, but with $(c,d) = (1,2)$ the outputs are $(1 \cdot 1, 0 \cdot 2) = (1,0)$ and $(2 \cdot 1, 1 \cdot 2) = (2,2)$, and $(1,0) \not\sim (2,2)$, since $1+2 \neq 0+2$. **And well defined is necessary, not sufficient.** The rule $[(a,b)] * [(c,d)] \coloneqq [(ad+bc,\ ac+bd)]$ is well defined, since it is the negative of the product; but it gives $[(0,1)] * [(0,1)] = [(0+0, 0+1)] = [(0,1)]$, so the "square" of the class of $(0,1)$ is that class itself, not $[(1,0)]$. Well-definedness says that a rule is a function; whether it is the right function is a separate question.

2.2.5 The natural numbers inside the integers

The integers were built to contain the natural numbers and more. What $\mathbb{Z}$ contains is a **copy** of $\mathbb{N}$: the classes in normal form $[(n,0)]$, each a set of pairs. Whether a natural number n is itself an element of $\mathbb{Z}$ depends on what the elements of a Peano system are, which the axioms leave open (Section 1.5); the copy is all we use, and it has the same arithmetic.

Proposition 2.2.3. The Natural Numbers Embed in the Integers

Let $\iota\colon \mathbb{N} \to \mathbb{Z}$, $\iota(n) \coloneqq [(n,0)]$.

(a) ι is injective, and for all $m, n \in \mathbb{N}$,

$$\iota(m+n) = \iota(m) + \iota(n), \qquad \iota(mn) = \iota(m)\,\iota(n).$$

(b) For every $x \in \mathbb{Z}$, $x + \iota(0) = x$ and $\iota(1) \cdot x = x$. For every $(a,b) \in \mathbb{N} \times \mathbb{N}$, $[(a,b)] = \iota(a) + (-\iota(b))$.

(c) $\mathbb{Z} = \iota(\mathbb{N}) \cup \{-\iota(n) : n \in \mathbb{N}\}$.

Idea. The identities in (a) and (b) are computations with the formulas of Definition 2.2.1 in which at least one operand is some $\iota(n) = [(n,0)]$, and its zero second coordinate makes most terms vanish. Injectivity and (c) are the two halves of Proposition 2.1.2: (b) of it says that different natural numbers give different classes, and (a) says that every class is in normal form.

Proof. *(a)* If $\iota(m) = \iota(n)$, then $[(m,0)] = [(n,0)]$, so $m = n$ by Proposition 2.1.2 (b); hence ι is injective (Sets and Functions, Definition 2.2.1). By Definition 2.2.1,

$$\iota(m) + \iota(n) = [(m+n,\ 0+0)] = [(m+n, 0)] = \iota(m+n),$$

since $0+0=0$ (Definition 1.2.1 (a)), and

$$\iota(m)\,\iota(n) = [(mn+0\cdot 0,\ m\cdot 0+0\cdot n)] = [(mn,0)] = \iota(mn),$$

since $m\cdot 0=0$ (Definition 1.2.1 (b)), $0\cdot n=0$ (Theorem 1.2.3 (e)) and $k+0=k$.

(b) Let $x=[(a,b)]$. Then $x+\iota(0)=[(a+0,b+0)]=[(a,b)]=x$, and

$$\iota(1)\cdot x = [(1\cdot a+0\cdot b,\ 1\cdot b+0\cdot a)] = [(a,b)] = x,$$

by Theorem 1.2.3 (e) ($1\cdot n=n$, $0\cdot n=0$) and $n+0=n$. Finally, $-\iota(b)=-[(b,0)]=[(0,b)]$, so $\iota(a)+(-\iota(b))=[(a+0,\ 0+b)]=[(a,b)]$, using $0+b=b$ (Theorem 1.2.3 (a)).

(c) ($\supseteq$) Both sets on the right consist of integers. ($\subseteq$) Let $x\in\mathbb{Z}$. By Proposition 2.1.2 (a), $x=[(k,0)]$ or $x=[(0,k)]$ for some $k\in\mathbb{N}$. In the first case $x=\iota(k)$. In the second, $x=[(0,k)]=-[(k,0)]=-\iota(k)$ by Definition 2.2.1. So x lies in the right side.

■

Identification. From now on we write n for $\iota(n)$ and regard $\mathbb{N}$ as a subset of $\mathbb{Z}$. This is a convenient abuse of notation, and (a) makes it honest: because ι is injective, different natural numbers stay different; because ι preserves $+$ and $\cdot$, a sum or product of natural numbers is the same whether computed in $\mathbb{N}$ or in $\mathbb{Z}$. One could rebuild $\mathbb{Z}$ so that it contains $\mathbb{N}$ literally, by putting each n in place of $\iota(n)$, but nothing later needs it. The order of $\mathbb{N}$ is matched to an order on $\mathbb{Z}$ in Section 2.4.

With this convention, (b) reads $[(a,b)]=a+(-b)$: the pair (a,b) records the difference it was meant to record. The notation $a-b$ is defined in Section 2.3, where $x-y:=x+(-y)$ is set up in every ring at once. Clause (c) reads

$$\mathbb{Z} = \mathbb{N}\cup\{-n : n\in\mathbb{N}\},$$

which is word for word the definition of the integers in *Sets and Functions*, Section 3.1. There it was a definition inside the familiar real numbers; here it is a theorem about a set built from $\mathbb{N}$. The positive integers keep the definition of *Sets and Functions*, Section 3.1,

$$\mathbb{Z}_{>0} := \{n\in\mathbb{N} : n\neq 0\},$$

which mentions only $\mathbb{N}$, and so is now a subset of $\mathbb{Z}$. In particular $5+x=3$ has the solution $x=[(0,2)]=-2$ in $\mathbb{Z}$, since $[(5,0)]+[(0,2)]=[(5,2)]=[(3,0)]$ by Proposition 2.1.2 (a).

Quick check. Compute $[(2,5)]\cdot[(4,1)]$ from Definition 2.2.1, and write the two factors and the product as n or $-n$ with $n\in\mathbb{N}$.

Solution. $[(2,5)]\cdot[(4,1)]=[(2\cdot 4+5\cdot 1,\ 2\cdot 1+5\cdot 4)]=[(13,22)]$. By Proposition 2.1.2 (a): $5=2+3$, so $[(2,5)]=[(0,3)]=-[(3,0)]=-3$ (the negative of Definition 2.2.1); $4=1+3$, so $[(4,1)]=[(3,0)]=3$; and $22=13+9$, so $[(13,22)]=[(0,9)]=-[(9,0)]=-9$. So the product of -3 and 3 is -9, as school predicts; here it is a computation in $\mathbb{N}$, not a rule of signs.

The integers now have a sum, a product and a negative, each a function on classes, and they

contain a copy of $\mathbb{N}$ with its arithmetic. Section 2.3 checks that these operations obey the usual laws, by reducing each law to a law of $\mathbb{N}$; Section 2.4 orders $\mathbb{Z}$.

2.2.6 Exercises

2.2.6.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Proposition 2.2.2.

(a) Which hypothesis of Theorem 2.2.1 (b) does the computation leading to Eq. 2.2.2 verify?

(b) Run that computation with $(a,b) = (1,3)$, $(a',b') = (4,6)$ and $(c,d) = (2,0)$: check that $(a,b) \sim (a',b')$, compute both outputs, and check that they are equivalent.

(c) Where in the proof are the laws of arithmetic in $\mathbb{N}$ used?

(d) Give the idea of the proof in two sentences.

Solution.

(a) The first: for all x, x', y, $x \sim x'$ implies $g.(x,y) \sim g.(x',y)$, with $x = (a,b)$, $x' = (a',b')$, $y = (c,d)$.

(b) $1 + 6 = 7 = 3 + 4$, so $(1,3) \sim (4,6)$. The outputs are

$$(ac + bd,\ ad + bc) = (2 + 0,\ 0 + 6) = (2,6),$$
$$(a'c + b'd,\ a'd + b'c) = (8 + 0,\ 0 + 12) = (8,12),$$

and $2 + 12 = 14 = 6 + 8$, so they are equivalent. The two intermediate equations are $ac + b'c = bc + a'c$, here $2 + 12 = 6 + 8$, and $bd + a'd = ad + b'd$, here $0 = 0$, since $d = 0$. Both outputs are representatives of -4, and indeed $(-2) \cdot 2 = -4$.

(c) In every rearrangement: commutativity and associativity of $+$ (to regroup the four-term sums), distributivity (to multiply Eq. 2.2.1 by c and by d), and commutativity of $\cdot$ and $+$ (in the second-argument step, to see that both formulas are symmetric in the two input pairs).

(d) A change of representative in one argument is an equation in $\mathbb{N}$, and multiplying it by c and by d and adding gives exactly the cross-sum equation for the two products. The second argument follows by symmetry of the formula, and transitivity (inside Theorem 2.2.1 (b)) combines the two changes.

2.2.6.2 B. Practice

Exercise. B1: Two rules that pass the check

Prove that each rule defines a function $\mathbb{Z} \to \mathbb{Z}$, and identify it in terms of the operations of Definition 2.2.1.

(a) $[(a,b)] \mapsto [(b,a)]$.

(b) $[(a,b)] \mapsto [(a+b,\ a+b)]$.

Solution.

(a) Let $(a,b) \sim (a',b')$, that is, $a+b' = b+a'$. The target $(b,a) \sim (b',a')$ is $b+a' = a+b'$, the same equation read from right to left. By Theorem 2.2.1 (a), the rule is a function. It is the negative $x \mapsto -x$ of Definition 2.2.1.

(b) Every output is the zero class: $(a+b, a+b) \sim (0,0)$, since $(a+b)+0 = (a+b)+0$. So for $(a,b) \sim (a',b')$ the outputs $(a+b, a+b)$ and $(a'+b', a'+b')$ are both related to $(0,0)$, hence to each other by (Eq2) and (Eq3) of Proposition 2.1.1, and Theorem 2.2.1 (a) applies. The function is the constant map $x \mapsto 0$, where $0 = \iota(0) = [(0,0)]$. In terms of Definition 2.2.1 it is $x \mapsto x+(-x)$, since $[(a,b)]+[(b,a)] = [(a+b,\ b+a)] = [(a+b,\ a+b)]$ by Theorem 1.2.3 (b).

Exercise. B2: Determine which

Determine which of the following rules define a function. Justify your answer.

(a) $\mathbb{Z} \to \mathbb{N}$, $[(a,b)] \mapsto a+b$.

(b) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(2a, 2b)]$.

(c) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(a^2, b^2)]$, where $n^2 := n \cdot n$.

Solution.

(a) No. $[(0,0)] = [(1,1)]$, since $0+1 = 0+1$, but the rule gives $0+0 = 0$ for the first representative and $1+1 = 2$ for the second.

(b) Yes. If $a+b' = b+a'$, then by distributivity (Theorem 1.2.3 (d))

$$2a+2b' = 2(a+b') = 2(b+a') = 2b+2a',$$

which is $(2a, 2b) \sim (2a', 2b')$; Theorem 2.2.1 (a) applies. The function is $x \mapsto x+x$: $[(a,b)]+[(a,b)] = [(a+a, b+b)]$, and $2n = (1+1)n = n+n$ by $1+1 = 2$ (Definition 1.2.1 (c)) and Theorem 1.2.3 (d), (e).

(c) No. $[(0,1)] = [(1,2)]$, since $0+2 = 1+1$. The rule gives $[(0,1)]$ for the first representative and $[(1,4)]$ for the second, and $(0,1) \not\sim (1,4)$, since $0+4 = 4 \neq 2 = 1+1$.

Exercise. B3: Prove or disprove

Prove or disprove that each rule defines a function.

(a) $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $([(a,b)], [(c,d)]) \mapsto [(a+d,\ b+c)]$.

(b) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(b,\ a+b)]$.

(c) $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $([(a,b)], [(c,d)]) \mapsto [(ad,\ bc)]$.

(d) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(a \cdot a + b \cdot b,\ a \cdot b + b \cdot a)]$.

Solution.

(a) **True.** The formula is not symmetric in the two input pairs, so both steps of Exam-

ple (Method: check on representatives, one argument at a time) are needed. *First argument:* if $a+b'=b+a'$, then

$$\begin{aligned}(a+d)+(b'+c) &= (a+b')+(c+d)\\ &= (b+a')+(c+d) = (b+c)+(a'+d),\end{aligned}$$

which is $(a+d,b+c)\sim(a'+d,b'+c)$. *Second argument:* if $c+d'=d+c'$, then

$$\begin{aligned}(a+d)+(b+c') &= (a+b)+(d+c')\\ &= (a+b)+(c+d') = (b+c)+(a+d'),\end{aligned}$$

which is $(a+d,b+c)\sim(a+d',b+c')$. The rearrangements are by Theorem 1.2.3 (b), (c), and Theorem 2.2.1 (b) concludes. The rule is $(x,y)\mapsto x+(-y)$, since $[(a,b)]+[(d,c)]=[(a+d,\ b+c)]$.

(b) **False.** $(0,0)\sim(1,1)$, but the outputs are $(0,0+0)=(0,0)$ and $(1,1+1)=(1,2)$, and $(0,0)\not\sim(1,2)$, since $0+2\neq 0+1$. The heuristic predicts the failure: $b-(a+b)=-a$ depends on a, not on $a-b$ alone.

(c) **False.** $(1,0)\sim(2,1)$, since $1+1=0+2$. With $(c,d)=(0,1)$ in the second argument, the outputs are $(1\cdot 1,\ 0\cdot 0)=(1,0)$ and $(2\cdot 1,\ 1\cdot 0)=(2,0)$, and $(1,0)\not\sim(2,0)$, since $1+0\neq 0+2$.

(d) **True.** The output is $g.(x,x)$ for $x=(a,b)$. If $x\sim x'$, then $g.(x,x)\sim g.(x',x)$ and $g.(x',x)\sim g.(x',x')$, by the two hypotheses checked in Proposition 2.2.2; transitivity, (Eq3) of Proposition 2.1.1, joins them, and Theorem 2.2.1 (a) applies. The rule is $x\mapsto x\cdot x$.

2.2.6.3 C. Going deeper

Exercise. C1: Properties of classes, and an operation that half works

(a) Prove Theorem 2.2.1 (c), and give an example, different from the one in the remark after the theorem, of a subset $S\subseteq\mathbb{N}\times\mathbb{N}$ for which its conclusion fails for $\sim$ of Definition 2.1.1.

(b) Give an example of an equivalence relation $\sim$ on $\mathbb{N}$ and a function $g\colon\mathbb{N}\times\mathbb{N}\to\mathbb{N}$ such that the first hypothesis of Theorem 2.2.1 (b) holds and the second fails. Check each property, and show that no function G as in the conclusion of (b) exists for your example.

Solution.

(a) *Proof.* Let $x\in X$. If $x\in S$, then $[x]\in\bar{S}$ by the definition of $\bar{S}$. Conversely, if $[x]\in\bar{S}$, there is $x'\in S$ with $[x]=[x']$; then $x\sim x'$ (Sets and Functions, Theorem 5.2.1 (b)), so $x'\sim x$ by symmetry, and the hypothesis on S, applied to $x'\sim x$ and $x'\in S$, gives $x\in S$.
Example. Let $S:=\{(a,b)\in\mathbb{N}\times\mathbb{N}:a=0\}$. Then $(0,0)\in S$, so $[(0,0)]\in\bar{S}$; and $[(1,1)]=[(0,0)]$, so $[(1,1)]\in\bar{S}$, but $(1,1)\notin S$. The hypothesis fails at $(0,0)\sim(1,1)$. Any example must contain a pair of S and an equivalent pair outside S, since that is exactly the negation of the hypothesis.

(b) *Example.* On $\mathbb{N}$, let $x\sim y$ mean "$x=0$ if and only if $y=0$". This is an equivalence relation: it says that x and y have the same value under the function sending 0 to 0

and every other number to 1, so it is an equivalence relation by Sets and Functions, Example (The fibers of a function). Its two classes are $\{0\}$ and $\mathbb{N} \setminus \{0\}$. Let

$$g(x, y) := \begin{cases} x & \text{if } y = 1, \\ 0 & \text{if } y \neq 1. \end{cases}$$

First hypothesis holds. Let $x \sim x'$ and $y \in \mathbb{N}$. If $y = 1$, then $g(x, y) = x \sim x' = g(x', y)$. If $y \neq 1$, both values are 0, and $0 \sim 0$.

Second hypothesis fails. $1 \sim 2$, since neither is 0; but $g(5, 1) = 5$ and $g(5, 2) = 0$, and $5 \not\sim 0$.

No induced G. If $G([x], [y]) = [g(x, y)]$ held for all x, y, then, since $[1] = [2]$, we would get $[5] = [g(5, 1)] = G([5], [1]) = G([5], [2]) = [g(5, 2)] = [0]$, and $[5] \neq [0]$ because $5 \not\sim 0$.

What any example must have. (i) $\sim$ is not equality: some class has two elements $y \sim y'$ with $y \neq y'$, else the second hypothesis holds trivially. (ii) $\sim$ has at least two classes, else every two outputs are related and both hypotheses hold. (iii) For some x and some $y \sim y'$, the values $g(x, y)$ and $g(x, y')$ lie in different classes, while for every y, changing x within its class never moves $g(x, y)$ out of its class. A different answer can be checked against (i)–(iii).

Exercise. C2: One at a time or both at once

Let $\sim$ be an equivalence relation on X and $g\colon X \times X \to X$ a function. Prove that the hypotheses of Theorem 2.2.1 (b) hold if and only if, for all $x, x', y, y' \in X$,

$$x \sim x' \text{ and } y \sim y' \implies g(x, y) \sim g(x', y').$$

Name the property of $\sim$ that each direction uses.

Solution. ($\Rightarrow$) Let $x \sim x'$ and $y \sim y'$. The first hypothesis, with y fixed, gives $g(x, y) \sim g(x', y)$; the second, with x' fixed, gives $g(x', y) \sim g(x', y')$. By **transitivity** (Eq3), $g(x, y) \sim g(x', y')$. This direction is the first paragraph of the proof of Theorem 2.2.1 (b).

($\Leftarrow$) Let $x \sim x'$ and $y \in X$. By **reflexivity** (Eq1), $y \sim y$, so the displayed condition with $y' = y$ gives $g(x, y) \sim g(x', y)$: the first hypothesis. Likewise $x \sim x$ and $y \sim y'$ give $g(x, y) \sim g(x, y')$: the second.

So the two forms are interchangeable. The one-at-a-time form is the one to check, because each check involves a single hypothesis $x \sim x'$; the both-at-once form is the one the proof of the lemma uses.